Final Practical – Multi-Stakeholder Presentation & Portfolio Guide

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Companion Presentation & Video Files in docs/:

- PowerPoint Presentation (.pptx): [FinDoc_AuditEngine_Presentation_Slides.pptx](./FinDoc_AuditEngine_Presentation_Slides.pptx) (12 Widescreen 16:9 Slides)
- Landscape Slide Deck (.pdf): [FinDoc_AuditEngine_Slide_Deck.pdf](./FinDoc_AuditEngine_Slide_Deck.pdf) (12 Designed 16:9 Slides)
- Recorded Demo Video (.mp4): [FinDoc_AuditEngine_Demo_Walkthrough.mp4](./FinDoc_AuditEngine_Demo_Walkthrough.mp4) (HD 720p Walkthrough)

Verified Automated Tests: Day 29: 28 passed (python -m pytest tests -q) | Day 30: 33 passed (python -m pytest tests -q) – *all collected tests passed*

1. Tailored Presentation Strategy by Audience (10–12 Minute Structure)

Track A: Presenting to Mentors

- Focus: Practical domain motivation, iterative improvements from code review through final polish, and honest engineering reflection.
- Key Points:

1. Inspired by real procurement workflows (managing purchase orders and vendor sponsorships for VIT's Riviera & Gravitas festivals).
2. Evolved into a clean, self-contained FastAPI working prototype with an institutional light-theme workbench (1 USD = 86.50 INR).
3. Demonstrates three representative scenarios clearly: a clean auto-approved invoice (INV-1001), a contract price discrepancy (INV-1004), and a high-risk anomaly (INV-1005).

Track B: Presenting to Team Members

- Focus: Code organization, API contracts, testing workflow, and reproducibility.
- Key Points:

1. Modular directory structure separating app/core (config, logging, regex privacy scrubber), app/services (3-way matcher, anomaly scorer, router), and app/routers (audit, mentor_prep, portfolio).
2. Fail-fast Pydantic v2 schema validators (@model_validator) enforcing $0.05 subtotal tolerances and valid GRN quantities.
3. Reproducible automated test suite (python -m pytest tests -q -> 28 passed in Day 29; 33 passed in Day 30 – all collected tests passed).

Track C: Presenting to the Technical Panel

- Focus: Mathematical logic, synthetic baseline scope, perturbation attribution, and transparent discussion of prototype limitations.
- Key Points:

1. Hybrid Design: Deterministic 3-way matching enforces configured SKU price/quantity rules, while the 7-feature IsolationForest surfaces unusual multi-feature patterns.
2. Accurate Model Scope: The IsolationForest is trained on a seeded synthetic baseline of 1,000 records (random_state=42), and its 0–100 output is an engineered anomaly-risk indicator rather than a calibrated probability of fraud.
3. Prototype Limitations & Future Work: Clearly explain that AUDIT_LEDGER is an in-memory Python list (requiring PostgreSQL for persistence), FinancialPrivacyScrubber and /parse-raw use regex rules (requiring NER and optical OCR for production), and model thresholds require real-world calibration.